ANNIHILATION BY AN ELEMENT PASSES FROM K TO K' ALONG A MAP ISOMORPHIC IN DEGREE ZERO AND SURJECTIVE IN DEGREE -1

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Allowed standard prerequisites (editorial).

- Hom(M[2], K') vanishes because M[2] has cohomology only in degrees <= -2 while K' has cohomology only in degrees -1 and 0. invoked at proof lines 22594-22596. Kills the factorization target, forcing f = 0 on K'.

Source reference. Original lemma 06XS: sources/stacks-chow/tags.txt, source lines 8861–8861. Cited at proof lines 22594-22596 for the vanishing of Hom(M[2], K'); the conclusion is used inside the counted span.

Lemma 0.2. *Let R be a ring. Let $\alpha : K \to K'$ be a morphism of $D(R)$. Assume*

(1) *$H^i(K) = H^i(K') = 0$ for $i \notin \{-1, 0\}$*
(2) *$H^0(\alpha)$ is an isomorphism and $H^{-1}(\alpha)$ is surjective.*

For any $f \in R$ if $f : K \to K$ is 0, then $f : K' \to K'$ is 0.

Proof. Set $M = \mathrm{Ker}(H^{-1}(\alpha))$. Then α fits into a distinguished triangle

$$M[1] \to K \to K' \to M[2]$$

Since $K \to K' \xrightarrow{f} K'$ is zero by our assumption, we see that $f : K' \to K'$ factors over a map $M[2] \to K'$. However $\mathrm{Hom}(M[2], K') = 0$ for example by Derived Categories, Lemma 06XS. □

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